

Project Executable Files

Date	11 July 2026
Team ID	xxxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	3 Marks

Project Executable Files Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example similar) or	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA

8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	No
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

VBCUA/

```

|
|— app.py
|— requirements.txt
|— README.md
|— .env.example
|— score.py
|— speech_processing.py
|— semantic_analysis.py
|— audio_features.py
|— report_generator.py
|
|— models/
|   |— whisper_model/
|   |— sentence_bert/
|
|— uploads/
|
|— reports/
|
|— assets/
|   |— logo.png
|   |— background.png
  
```

```

|   └─ icons/
|
|   └─ utils/
|       └─ helper.py
|       └─ preprocessing.py
|       └─ constants.py
|
└─ sample_audio/
    └─ sample1.wav
    └─ sample2.wav
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted/ Deployed URL	Not Depolyed
Login Credential s (Demo)	Not required
Platform / Hosting Provider	Local Streamlit Application
Repositor y Link	https://github.com/poshithayagna2007-max/VBCUA
Demo Video Link	https://drive.google.com/file/d/1rMGxhxHG_NIjDjdrAuU6Kl2C0na4Gl5K/view?usp=sharing

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

1. Download or clone the project repository.
2. Open the project folder in Visual Studio Code.
3. Create and activate a Python virtual environment.
4. Install all required dependencies using:

```
pip install -r requirements.txt
```

5. Configure the environment variables using the .env.example file.
6. Start the Streamlit application:

```
streamlit run app.py
```

7. The application will automatically open in the default web browser.
8. Upload an audio recording or record speech through the application.
9. Enter the expected concept/topic.
10. Click Evaluate to generate the Concept Understanding Score and download the evaluation report.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Requires internet connection for AI model/API services.	Stable internet connection required.
2	Performance may reduce with noisy audio recordings.	Use a clear microphone in a quiet environment.
3	Best accuracy achieved with English speech.	Future versions will support additional languages.